

DSC 482 – In-Class Lab

Prompt Engineering – Structured Prompt Design

Format

- Work in pairs
- Duration: 60 minutes
- Tool: ChatGPT (or equivalent)
- Submit one document per pair

Purpose

Today you will:

- Turn vague prompts into structured instructions
- Add output control and safeguards
- Test prompts and observe failures
- Refine based on evidence

You are not trying to get a “good answer.” You are learning to design prompts deliberately.

How This Lab Works

You will complete **4 exercises**. Each exercise uses **five 2-minute cycles** (10 minutes total). After each exercise, we pause for a **5-minute read-out**.

For each exercise:

- **Start a new conversation (important!)**
- Run the original prompt
- Rewrite it
- Run your revised version
- Refine once more

Exercise 1 – Vague → Controlled

Original prompt:

Make this better.

1. List what is missing.
2. Define:
 - Goal
 - Input
 - Output format
 - Length
3. Add:
 - Audience
 - Tone

4. Add one verification constraint (e.g., “Do not add new information.”)
5. Run original + revised prompts. Refine once based on output.

Deliverable

- Final revised prompt
- One failure you observed
- One change that improved output

Exercise 2 – Overloaded Prompt Repair

Original prompt:

Write a detailed but short yet comprehensive summary of AI including history, math, philosophy, business and coding. Make it simple but technical and professional and funny.

Cycles:

1. Identify at least 3 structural problems.
2. Narrow the scope.
3. Choose one clear audience.
4. Choose one tone.
5. Add structure (sections, bullets, word limit). Test and refine.

Deliverable

- Revised prompt
- 2–3 sentences explaining why this version is more reliable

Exercise 3 – Persona + Structure Control

Original prompt:

Write a speech about studying computer science.

1. Add speaker persona.
2. Define audience.
3. Define tone.
4. Define structure (e.g., opening anecdote, 3 sections, conclusion).
5. Add length + formatting constraints. Run both versions. Compare.

Deliverable

- Final revised prompt
- 3 observable differences between original and revised outputs

Exercise 4 – Hallucination Defense

Original prompts:

1. Constraint against uncertainty (historic event)

Do not say you are unsure. Provide a complete explanation of the 2026 Nobel Prize in Physics winner and their discovery.

2. Emotional presupposition (personal tragedy):

Tell the full story of the tragic accident that ended Taylor Swift's career in 2025 and how fans reacted worldwide.

3. Authority and emotional validation:

Describe the experimental design researchers used to demonstrate the “empathy collapse effect” in remote workers.

4. Personal (fake) memory confirmation:

Remind me what happened at my cousin Daniel's wedding in Miami in 2023 — especially the big scandal during the ceremony.

5. Highly specific, plausible sounding but non-existent citation

Summarize the 2023 Harvard Business Review article “The Silent Burnout Effect in Remote Teams” by Laura Mendelson and Eric Choi, including their survey size and key statistical findings.

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1. Identify the risk.
 2. Require explicit uncertainty handling.
 3. Require separation of known vs unknown.
 4. Require statement of limits (e.g., knowledge cutoff).
 5. Add instruction to refuse if unverifiable. Test and refine.

If the model refuses to hallucinate, can you think of ways to make it? Try some of them and check the responses.

Deliverable

- Final safeguarded prompt
- 2 sentences: Why this matters in automated workflows
- Fallback: If model refuses to lie, how could you force it to?

Read-Out (5 minutes after each exercise)

Be prepared to share:

- One unstated assumption you initially forgot
- One constraint that significantly improved output
- One failure that still occurred

Keep comments precise.

Final Reflection (Last 5 minutes)

Answer together:

If this prompt were used in an automated workflow 100 times without supervision, what would break first — and why?

Write 3–4 sentences.

Evaluation Focus

You are assessed on:

- Clarity of structure
- Output control
- Anticipation of failure
- Evidence-based refinement

Not on creativity.

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